

B30801 Managerial Accounting Autumn 2025

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Session #5: Balanced scorecard, non-financial performance, performance evaluation; Summary and MA toolbox.

CHICAGO BOOTH



The University of Chicago Booth School of Business

Balanced Scorecard and non-financial performance measurement

Why is this important?

- It seems obvious that non-financial performance measures – like customer loyalty and employee satisfaction – are important for long-term value creation.

Discounted Cash Flow (DCF)

$$V_0 = \sum_{t=1}^{\infty} \frac{CF_t}{(1+r)^t}$$

In the long-run, good things must happen (higher revenues, profits, and free cash flows) if customers and employees are loyal, satisfied, and happy!!

Why is this important?

- It ***seems obvious*** that non-financial performance measures – like customer loyalty and employee satisfaction – are important for long-term value creation.
- This is the idea behind a balanced scorecard – we need to communicate expectations, then measure, evaluate, and reward based on these measures.

Balanced Scorecard

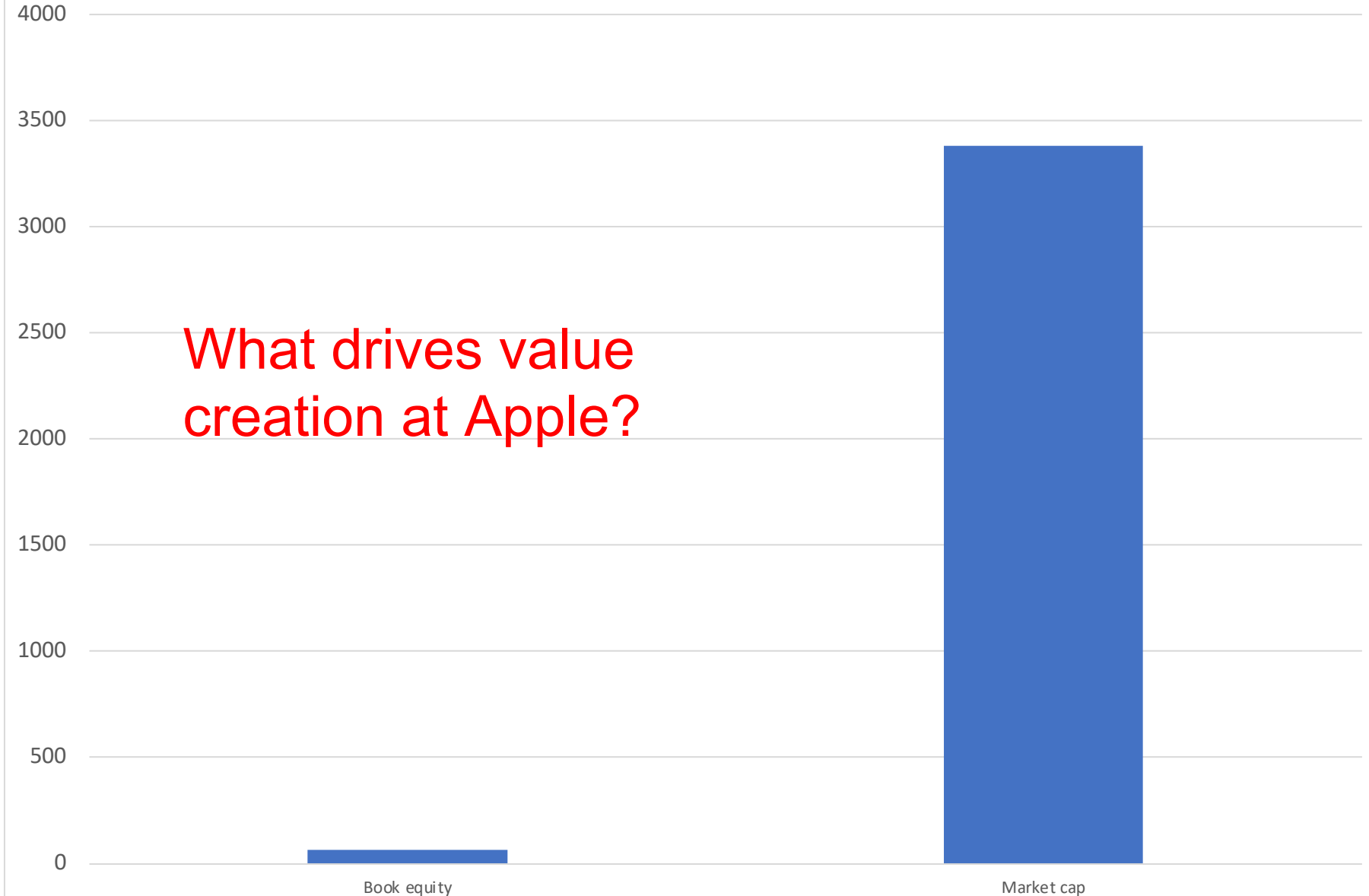
Four elements:

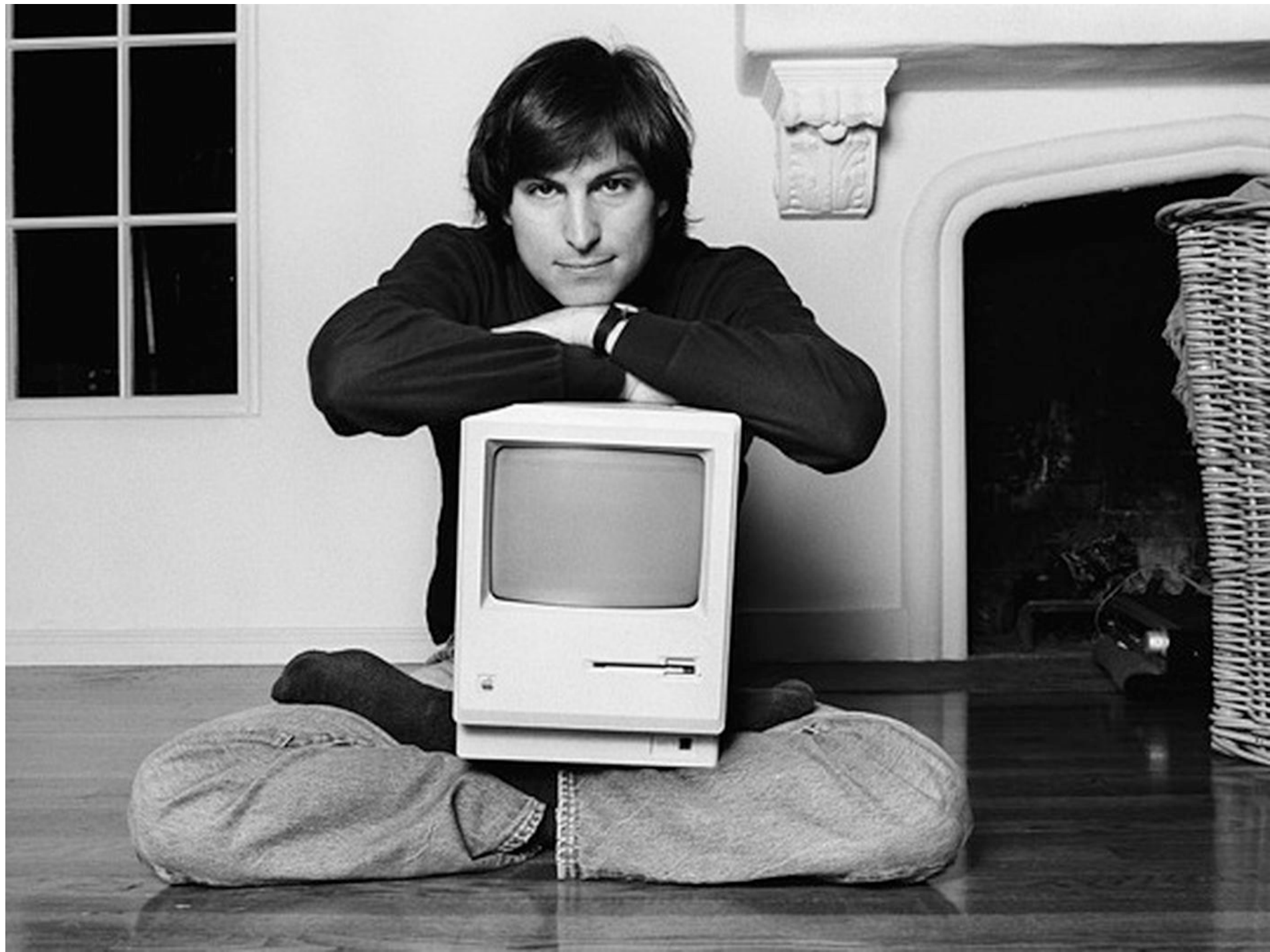
1. **Financial:** how is success measured by our shareholders?
2. **Customer:** how do we create value for our customers?
3. **Process:** which processes are important to meet expectations of customers and shareholders? What do we need to do well to succeed?
4. **Learning, growth, and innovation:** what employee, organizational, and systems capabilities do we need to continually improve our processes and customer relationships?

What is the motivation for non-financial performance measurement?

- GAAP-based accounting profitability measures not well-suited for today's economy, especially for bio-pharma and technology companies that make big investments in brand, intellectual property, technology (AI), often through spending on advertising, marketing, and R&D that is immediately expensed under GAAP.
- ABC is a response to this, but we can also measure, evaluate, and reward non-financial measures (e.g., customer satisfaction, which drives brand value).

AAPL: Book value vs. market cap. (\$M, at 8.22.25)











What non-financial measures drive value creation for brands?

What about product innovation?

What about AI?

Sources of Sustainable Competitive Advantage (ROIC > WACC)

Price premium

Innovative products: Difficult-to-copy or patented products, services, or technologies

Quality: Customers willing to pay a premium for a real or perceived difference in quality over and above competing products or services

Brand: Customers willing to pay a premium based on brand, even if there is no clear quality difference

Customer lock-in: Customers unwilling or unable to replace a product or service they use with a competing product or service

Rational price discipline: Lower bound on prices established by large industry leaders through price signaling or capacity management

Cost and capital efficiency

Innovative business method: Difficult-to-copy business method that contrasts with established industry practice

Unique resources: Advantage resulting from inherent geological characteristics or unique access to raw material(s)

Economies of scale: Efficient scale or size for the relevant market

Scalable product/process: Ability to add customers and capacity at negligible marginal cost

Network effects: Products or services that offer increasing value to customer with scale, at negligible or decreasing costs to scale, possibly combined with customer switching costs

What is the motivation for non-financial performance measurement?

1. Desirable non-financial outcomes—like higher customer satisfaction—ultimately (over the long-term) drive increases in long-run value. That is, non-financial measures are **leading indicators** of future financial performance.
 - We need to reward/compensate people today and measure non-financial performance now (we cannot wait several years).
 - Financial performance may not be measurable at lower levels of the organization.

What is the motivation for non-financial performance measurement?

2. Non-financial goals are part of the entity's mission or strategy:
 - Some are sector-specific: e.g., mine safety for natural resources firms; regulatory compliance and reputation for banks.
 - Some are more general: reduce carbon emissions.
 - Some entities only have non-financial goals (non-profits; government entities; but still a need for financial discipline/cash flow management).

Having decided that, as an entity, you do care about customer satisfaction (as one example), you have to build it into the Rules of the Game:

- You have to **allocate decision rights (resources)** to provide and manage resources to execute on the goal.
- You have to **measure** customer satisfaction, which is likely to have different elements.
- You have to **evaluate** this performance: weight it in the performance measurement system and link it to **compensation** (evaluate and reward).

The Best and Worst Airlines of 2023

U.S. carriers avoided many of the snafus of 2021 and 2022—here's who did the best job of delivering what fliers needed

By *Dawn Gilbertson* [Follow](#), *Allison Pohle* [Follow](#) and *Kevin McAllister* [Follow](#)

Jan. 24, 2024 5:30 am ET

On airlines' long, uneven shift from empty flights to packed planes, 2023 will go down as the year they got their act together. Mostly.

Major U.S. airlines sharply reduced cancellations and improved on-time arrivals, a relief for passengers after big-time travel troubles in 2021 and 2022. Hold the confetti cannons, though: Baggage handling and tarmac delays remained iffy, and passenger complaints were stubbornly high.

Anyone who boarded a flight likely felt a sense of order not seen since 2019. Still, better is far from perfect as the industry faces air-traffic control issues, clogged airports, weather woes and unforced errors.

And the winner is



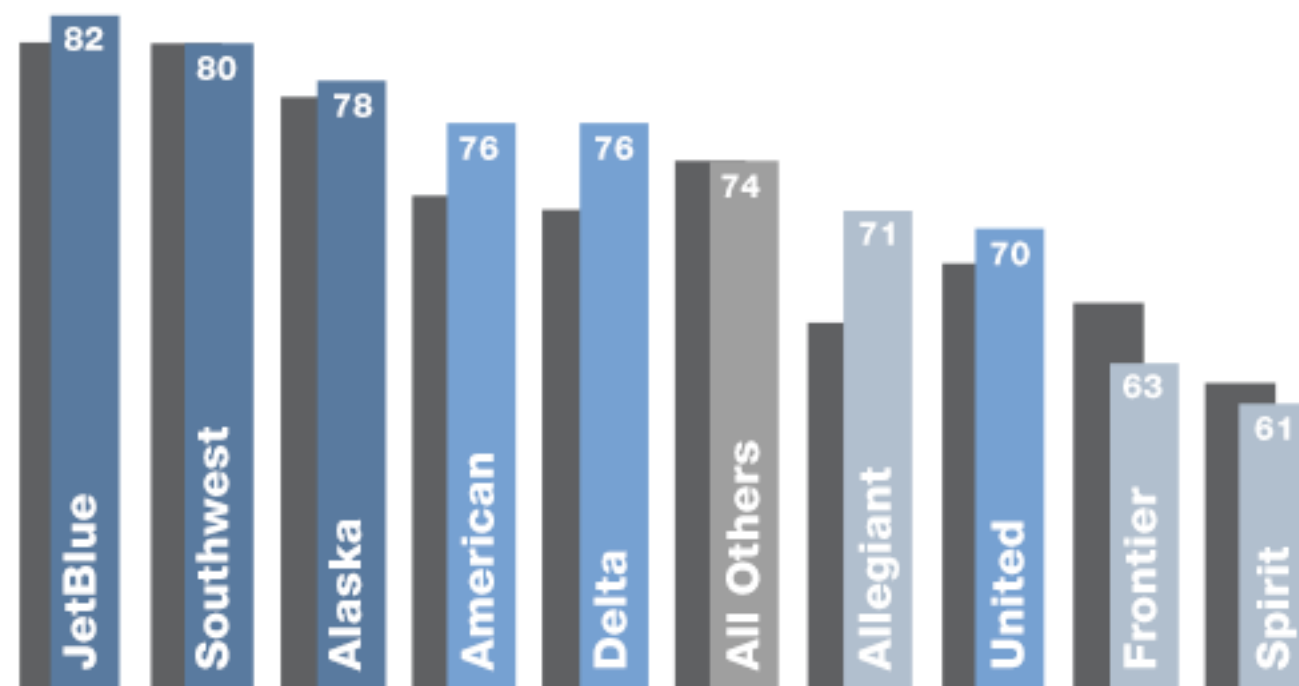
Delta Air Lines placed first in the rankings for the sixth time in seven years. PHOTO: TAIDGH BARRON/ZUMA PRESS WIRE

How do we measure customer satisfaction?

There are **multiple constructs**, each with **different elements**. For an airline we might have:

- Check-in experience.
- Boarding and gate experience.
- On-time performance; cancellation frequency.
- In-flight experience (meals, service, seat comfort).
- Reliability of baggage handling.
- Flexibility of ticketing.
- Frequent-flier program.
- Management of adverse situations; customer service.

Airlines by Category ACSI: Year-Over-Year





If you're United (for example), do you care if you have low customer satisfaction if you can sell all available seats?

X close

Print

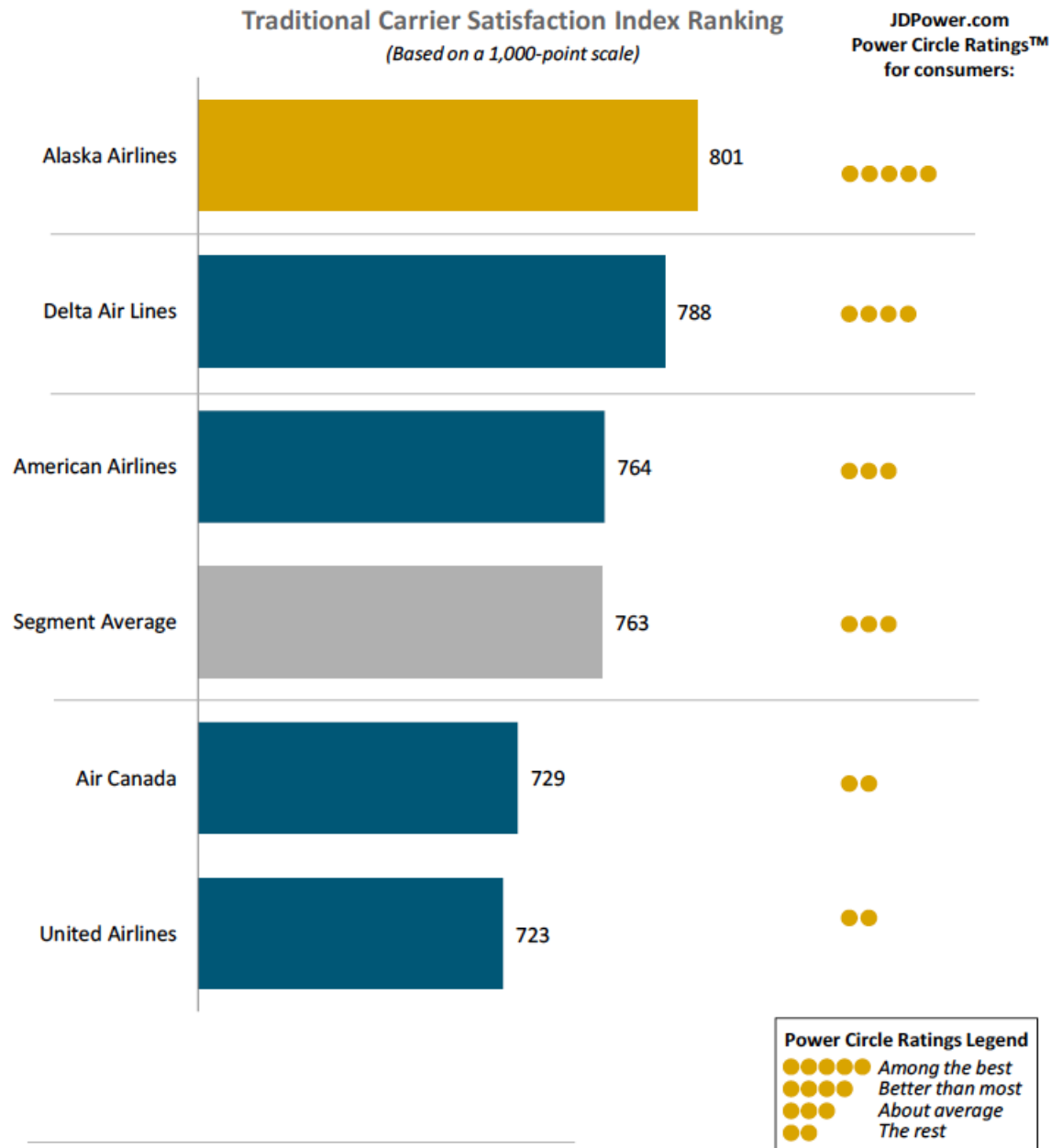
Benchmarks By Industry

Airlines

	Base-line	95	96	97	98	99	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	Previous Year % Change	First Year % Change
JetBlue	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	81	83	79	81	80	82	2.5	1.2
Southwest	78	76	76	76	74	72	70	70	74	75	73	74	74	76	79	81	79	81	77	81	78	78	80	80	0.0	2.6
Alaska	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	75	77	78	1.3	4.0
American	70	71	71	62	67	64	63	62	63	67	66	64	62	60	62	60	63	63	64	65	66	66	72	76	5.6	8.6
Delta	77	72	67	69	65	68	66	61	66	67	67	65	64	59	60	64	62	56	65	68	71	71	71	76	7.0	-1.3
Airlines	72	69	69	67	65	63	63	61	66	67	66	66	65	63	62	64	66	65	67	69	69	69	72	75	4.2	4.2
All Others	NM	70	74	70	62	67	63	64	72	74	73	74	74	75	75	77	75	76	74	72	70	73	74	74	0.0	5.7
Allegiant	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	65	65	71	9.2	9.2
United	71	67	70	68	65	62	62	59	64	63	64	61	63	56	56	56	60	61	62	62	60	60	68	70	2.9	-1.4
Frontier	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	58	66	63	-4.5	8.6
Spirit	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM	54	62	61	-1.6	13.0
US Airways	72	67	66	68	65	61	62	60	63	64	62	57	62	61	54	59	62	61	65	64	66	#			N/A	N/A
Northwest Airlines	69	71	67	64	63	53	62	56	65	64	64	64	61	61	57	57	61	#							N/A	N/A
Continental	67	64	66	64	66	64	62	67	68	68	67	70	67	69	62	68	71	64	#						N/A	N/A

J.D. Power

2019 North America Airline Satisfaction StudySM



Source: J.D. Power 2019 North America Airline Satisfaction StudySM



The Airline Quality Rating (AQR) is the premier statistical study of major airline performance in the United States, objectively comparing airline quality.

2018 AIRLINE RANKINGS

The **Airline Quality Rating (AQR)** is the most comprehensive study of performance and quality of the largest airlines in the United States. The rating is a multifactor examination of the airlines based on mishandled baggage, consumer complaints, on-time performance and involuntary denied boardings.

The Airline Quality Rating industry score for 2017 shows an industry that improved in overall performance quality over the previous year. Nine airlines (American, ExpressJet, Frontier, Hawaiian, JetBlue, SkyWest, Southwest, Spirit and United) showed improvement in AQR scores in 2017.

Frontier had the largest improvement in their AQR score in 2017. Three airlines (Alaska, Delta and Virgin America) all had a decline in their 2017 AQR

- | | |
|---|--------------------|
| 1 | Alaska Airlines |
| 2 | Delta Air Lines |
| 3 | JetBlue |
| 4 | Hawaiian Airlines |
| 5 | Southwest Airlines |
| 6 | SkyWest Airlines |
| 7 | Virgin America |
| 8 | United Airlines |
| 9 | American Airlines |

The industry **mishandled baggage** rate was better, decreasing from 2.70 per 1,000 passengers in 2016 to **2.46** per 1,000 passengers in 2017.

Involuntary **denied boardings** by the industry improved to **0.34** per 10,000 passengers in 2017 from 0.62 per 10,000 passengers in 2016.

The **consumer complaint** rate across the industry declined to **1.35** per 100,000 passengers in 2017 from 1.52 per 100,000 passengers in 2016. Of the 11,570 complaints registered with the DOT regarding all U.S. domestic carriers, 74% were for flight problems, baggage problems, reservation, ticketing and boarding issues, or customer service problems. The 10% decrease in the rate of consumer complaints in 2017 suggests that

TABLE 1. OVERALL PERCENTAGE OF REPORTED FLIGHT OPERATIONS ARRIVING ON TIME BY CARRIER*

APRIL 2018

	AT 30 LARGEST AIRPORTS		AT ALL US AIRPORTS	
CARRIER*	NUMBER OF AIRPORTS REPORTED	PERCENT OF ARRIVALS ON TIME	NUMBER OF AIRPORTS REPORTED	PERCENT OF ARRIVALS ON TIME
HAWAIIAN AIRLINES	9	85.0	17	87.7
DELTA AIR LINES	30	85.8	147	86.4
MESA AIRLINES	15	85.4	100	85.1
AMERICAN AIRLINES	29	84.0	98	83.9
UNITED AIRLINES	28	83.8	101	83.9
ALASKA AIRLINES**	26	82.0	70	83.4
SKYWEST AIRLINES	26	82.8	227	83.0
SPIRIT AIRLINES	21	81.2	41	81.8
ENVOY AIRLINES	15	80.0	132	81.1
EXPRESSJET AIRLINES	14	78.8	118	80.9
REPUBLIC AIRLINE	17	79.4	82	80.3
PSA AIRLINES	11	81.3	96	79.4
ALLEGiant AIRLINES	7	77.7	118	78.8
ENDEAVOR AIRLINES	17	77.9	104	78.1
SOUTHWEST AIRLINES	24	78.0	86	77.7
FRONTIER AIRLINES	23	75.9	70	76.4
JETBLUE AIRWAYS	25	67.7	64	67.6
TOTAL		81.5		81.3

Source: US Department of Transportation, Air Travel Consumer Report June 2018,

TABLE 8. OVERALL NUMBER AND PERCENTAGE OF FLIGHT CANCELLATIONS BY CARRIER

APRIL 2018

CARRIER	AT 30 LARGEST U.S. AIRPORTS B/				AT ALL US AIRPORTS C/			
	NUMBER OF AIRPORTS REPORTED	FLIGHT OPERATIONS SCHEDULED	FLIGHT OPERATIONS CANCELLED	PERCENT OF OPERATIONS CANCELLED	NUMBER OF AIRPORTS REPORTED	FLIGHT OPERATIONS SCHEDULED	FLIGHT OPERATIONS CANCELLED	PERCENT OF OPERATIONS CANCELLED
ENDEAVOR	17	11574	305	2.6	104	20901	541	2.6
REPUBLIC	17	16630	483	2.9	82	26219	660	2.5
ENVOY	15	12721	326	2.6	132	24068	580	2.4
PSA	11	12192	215	1.8	96	23303	454	1.9
EXPRESSJET	14	9088	134	1.5	118	17644	243	1.4
SKYWEST	26	34617	437	1.3	227	62861	830	1.3
JETBLUE	25	18140	220	1.2	64	25836	331	1.3
MESA	15	9850	116	1.2	100	16726	214	1.3
SOUTHWEST	24	60009	673	1.1	86	114117	1220	1.1
FRONTIER	23	5892	44	0.7	70	9615	94	1.0
SPIRIT	21	11572	81	0.7	41	14548	100	0.7
AMERICAN	29	60600	360	0.6	98	74384	456	0.6
ALASKA*	26	15886	99	0.6	70	21987	127	0.6
ALLEGIAN	7	1207	1	0.1	118	8146	30	0.4
DELTA	30	59782	203	0.3	147	79013	268	0.3
UNITED	28	41995	78	0.2	101	49937	94	0.2
HAWAIIAN	9	3344	3	0.1	17	6741	10	0.1
TOTAL		385099	3778	1.0		596046	6252	1.0

**MISHANDLED BAGGAGE REPORTS FILED BY PASSENGERS
U.S. AIRLINES***

		APRIL 2018			APRIL 2017		
RANK	AIRLINE	TOTAL BAGGAGE REPORTS	ENPLANED PASSENGERS	REPORTS PER 1000 PASSENGERS	TOTAL BAGGAGE REPORTS	ENPLANED PASSENGERS	REPORTS PER 1000 PASSENGERS
1	DELTA AIR LINES	15,231	9,975,059	1.53	30,282	9,953,754	3.04
2	SPIRIT AIRLINES	3,204	2,090,046	1.53	2,656	1,813,630	1.46
3	JETBLUE AIRWAYS	4,573	2,867,720	1.59	4,377	2,913,464	1.50
4	UNITED AIRLINES	14,418	7,269,153	1.98	14,249	6,734,304	2.12
5	ALASKA AIRLINES**	6,453	2,912,037	2.22	2,838	2,008,750	1.41
6	FRONTIER AIRLINES	3,739	1,510,781	2.47	2,881	1,248,519	2.31
7	SOUTHWEST AIRLINES	34,852	13,605,528	2.56	32,361	13,331,080	2.43
8	HAWAIIAN AIRLINES	2,335	837,744	2.79	2,068	819,181	2.52
9	AMERICAN AIRLINES	28,097	9,205,782	3.05	28,327	10,070,814	2.81
10	SKYWEST AIRLINES	9,671	2,472,549	3.91	8,446	2,783,144	3.03
11	EXPRESSJET AIRLINES	2,790	695,995	4.01	6,179	1,324,149	4.67
12	ENVOY AIR	4,912	993,029	4.95	-	-	-
	TOTALS	130,275	54,435,423	2.39	134,664	53,000,789	2.54

**PASSENGERS DENIED BOARDING
BY U.S. AIRLINES***

Rank	Airline	JANUARY - MARCH 2018				JANUARY - MARCH 2017			
		<u>DENIED BOARDINGS (DB'S)</u>		Enplaned Passengers	Involuntary DB's per 10,000 Passengers	<u>DENIED BOARDINGS (DB'S)</u>		Enplaned Passengers	Involuntary DB's per 10,000 Passengers
		Voluntary	Involuntary			Voluntary	Involuntary		
1	DELTA AIR LINES	23,777	13	30,868,044	0.00	34,388	354	29,863,259	0.12
2	HAWAIIAN AIRLINES	160	2	2,676,265	0.01	113	55	2,649,691	0.21
3	JETBLUE AIRWAYS	844	7	8,927,623	0.01	553	1,415	8,770,054	1.61
4	EXPRESSJET AIRLINES	2,230	3	2,411,261	0.01	7,354	507	4,137,528	1.23
5	UNITED AIRLINES	8,214	27	21,314,280	0.01	15,917	900	20,559,648	0.44
6	ENDEAVOR AIR	3,117	4	2,709,157	0.01	-	-	-	-
7	PSA AIRLINES	2,416	22	3,210,770	0.07	-	-	-	-
8	REPUBLIC AIRLINE	2,613	32	4,050,777	0.08	-	-	-	-
9	VIRGIN AMERICA	89	22	2,045,185	0.11	908	51	1,803,849	0.28
10	SKYWEST AIRLINES	7,816	88	8,060,076	0.11	11,543	622	7,201,623	0.86
11	AMERICAN AIRLINES	15,658	483	31,525,870	0.15	10,870	2,301	30,582,875	0.75
12	MESA AIRLINES	2,413	47	2,864,946	0.16	-	-	-	-
13	ALLEGiant AIR	0	58	3,306,693	0.18	-	-	-	-
14	SOUTHWEST AIRLINES	4,325	669	37,042,370	0.18	16,205	2,537	35,246,083	0.72
15	ENVOY AIR	3,839	59	2,943,408	0.20	-	-	-	-
16	ALASKA AIRLINES	1,206	120	5,844,254	0.21	1,981	206	5,595,050	0.37
17	FRONTIER AIRLINES	949	188	4,416,868	0.43	312	167	3,582,185	0.47
18	SPIRIT AIRLINES	5,941	410	6,180,877	0.66	2,141	451	5,274,512	0.86
	TOTALS	85,607	2,254	180,398,724	0.12	102,285	9,566	155,266,357	0.62

Improving customer satisfaction is an investment

Like any other investment, we need to evaluate whether improving customer satisfaction increases value.

Benefits:

- Greater loyalty
- Lower price elasticity
- Lower marketing costs (word of mouth, social media)

Improving customer satisfaction is an investment

Costs:

- Depends on attributes of products and services
- Costs often increase in a non-linear (convex) fashion; much harder to go from 90% → 100% than from 60% → 70%; do we really need to get to 100% satisfaction?
Conversely, the marginal benefits decrease as satisfaction levels increase.

Another problem – if you weight these measures in measuring and rewarding performance, people will look to game the system, which can be counter-productive:

EDUCATION

Temple University Fires a Dean Over Falsified M.B.A. Rankings Data

Temple asked Moshe Porat, its business school dean, to step down over inflated student data for a ranking of online M.B.A. programs



Temple University asked Fox Business School Dean Moshe Porat, seen in a 2009 photo, to step down this week after an investigation found his staff inflated student scores and grades sent to U.S. News & World Report for its ranking of online M.B.A. programs. PHOTO: APRIL SAUL/ASSOCIATED PRESS

By Kelsey Gee and Janaki Chadha

Updated July 13, 2018 2:49 p.m. ET

The dean of Temple University's business school was ousted this week after an outside investigation found employees had inflated student test scores and grade point averages used in calculating a closely watched M.B.A. ranking—a brewing controversy that triggered a request Friday by the Pennsylvania attorney general for a state probe.

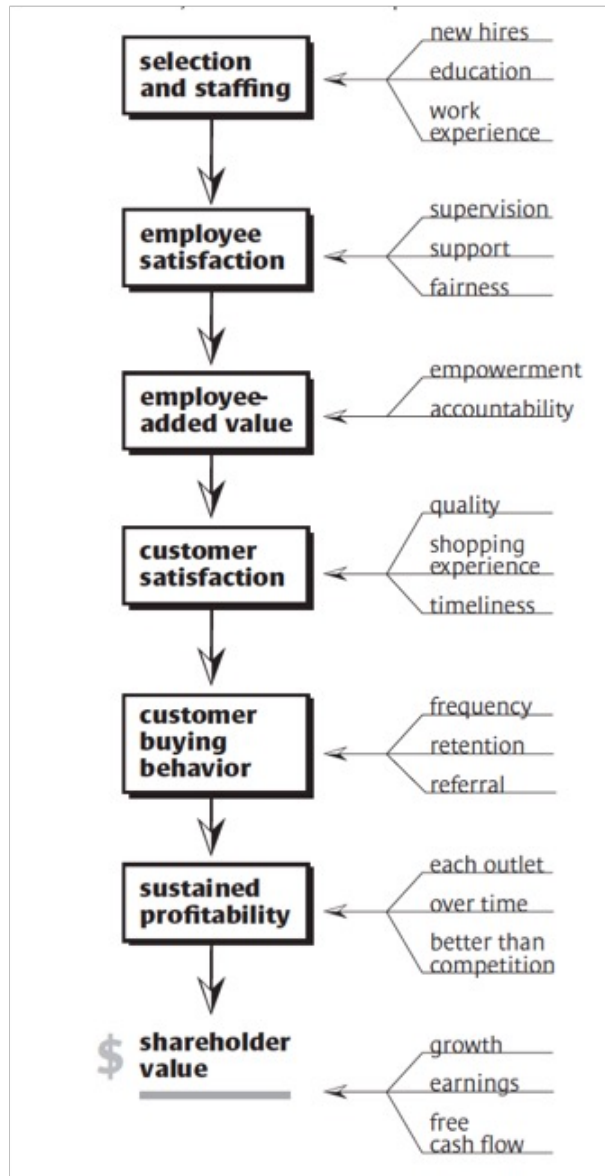
Temple's Fox School of Business held the top spot in a U.S. News and World Report list of over 200 online master's in business administration programs for four consecutive years, including the 2018 rankings. In January, the publication revised the school's position to "unranked" after being notified by Fox of its inaccuracies. The rankings are based on universities' self-reported data about the number of students submitting standardized test scores to gain admission and other factors, such as average debt held by its graduates.

The FT now asks business schools for data on how many hours of ESG they teach in their programs – this is an input to their business school rankings.

How to incorporate non-financial goals

- Develop and test **causal** models that link all of the factors that drive success, which is ultimately measured in financial terms.
- Modeling allows entities to identify and weight underlying drivers of performance.
- Evidence suggests that there are threshold effects and non-linearities:
 - Benefits begin at CS levels of 60% and don't increase beyond 75% to 80%.
 - Costs are increasing and convex in satisfaction—it's more costly to get from 80% to 90% than from 50% to 60%.

This is a “value driver map” or **causal model** from a fast food company (Ittner and Larcker, HBR).



The causal mechanisms matter:

- Employee satisfaction leads to higher retention of supervisory-level employees, which leads to higher quality and timeliness of food/delivery, which leads to higher customer satisfaction and so higher customer retention and referral which ultimately leads to higher sales and profits.
- Because retention of lower-level employees did not lead to better outcomes, focus investments on retention of supervisory-level people.



This is the ACSI customer satisfaction model. Customer interviews are an input to a multi-equation “cause-and-effect” econometric model. This model features:

- Indices for the drivers of satisfaction (light blue),
- Customer satisfaction (center),
- Outcomes on the right.
- Arrows represent causal “impacts”

The variables are measured using responses to survey questions (i.e., several questions are usually used to measure a given construct). Need to assess construct validity—how do we know variables are reliably measured?

So what do we do?

- Start with the objectives of the organization, as well as concrete strategy. Make sure non-financial measures map into the strategy/objectives.
- Develop **causal model**:
 - Underlying economic model of linkages (“value driver map”).
 - Define and measure variables, collect data.
 - Estimate model econometrically: goal is to establish **causal relationships**.

Key Takeaways

Direct cost; no ambiguity to assign it to a cost object.

ID cost; need to use some form of allocations. Profit margin with IDC, need to ask ourselves if the allocation makes sense, else the profit measure is inaccurate.

Allocation;

RegionFly and VMD MIC cases

- Direct and indirect (joint and common) costs.
- Indirect costs must be *allocated* to cost objects.
- Profit margins used to evaluate/compare products.
- Can lead to bad decisions.
 - Indirect cost allocations *inherently arbitrary*.
 - Why allocate costs at all?
 - Fundamentally depends on purpose – there are multiple purposes; different numbers (and cost allocations) for each.

Midwest Office Products (ABC)

ABC: Measure costs right and make the right decisions

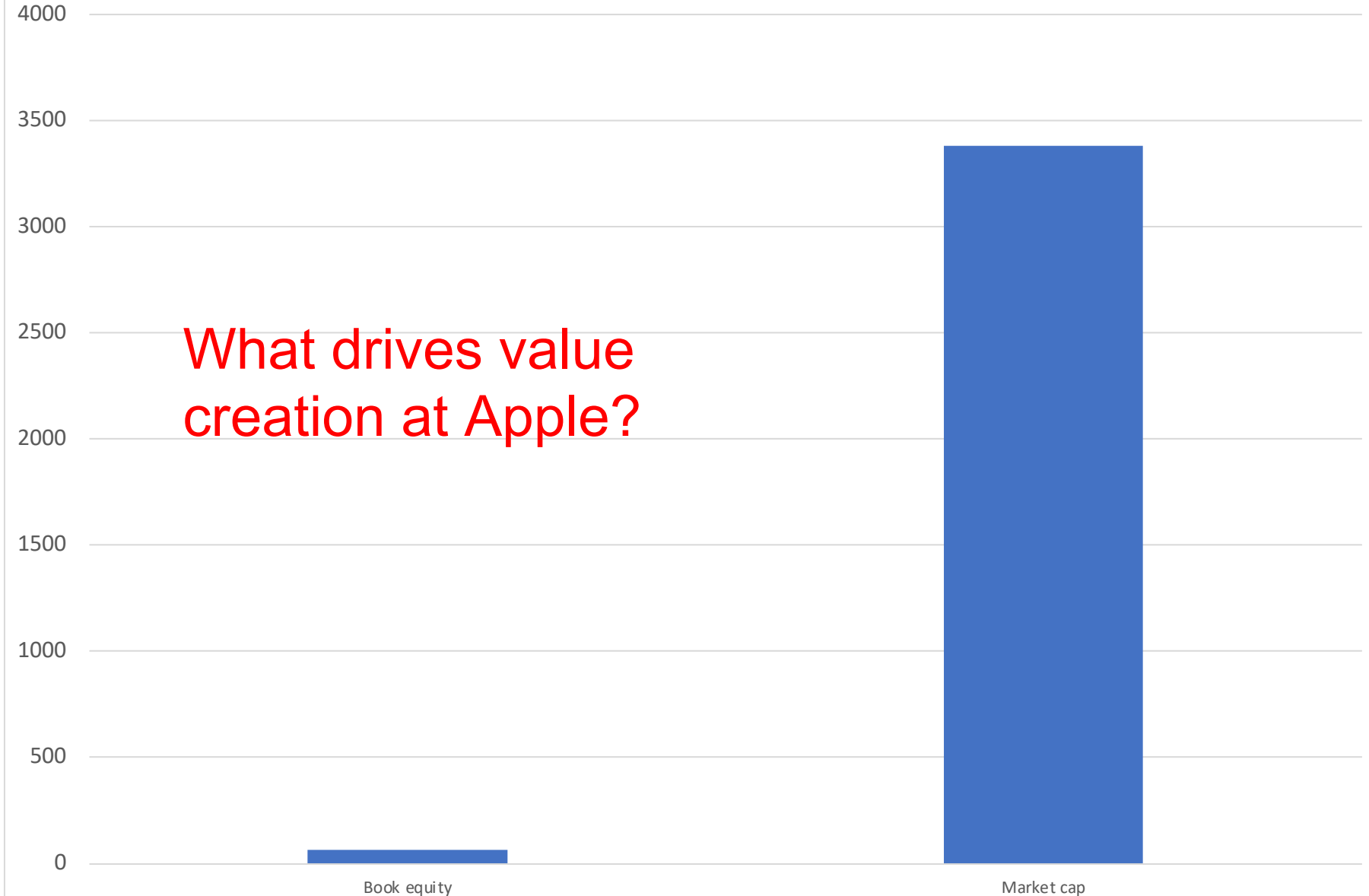
- What activities/processes drive value? What are the associated costs (resources consumed)?
- How to measure capacity (use *practical capacity*).
- Figure out cost drivers and cost of capacity.
- Allocate costs to activities based on use.
- Evaluate value proposition/make decision.

Birch Paper and Citibank PE cases: Rules of the game

Birch Paper: All about **coordination** and **control** – how to design rules of the game (how transfer prices are defined and used) to align internal management decisions with overall goals of organization.

Ask; what is the second best outcome? cos in practice and in a real org, we can rarely get to best / optimal outcome.

AAPL: Book value vs. market cap. (\$M, at 8.22.25)



Birch and Citibank PE → Rules of the game

Citi PE: Non-financial performance measurement

- Why include non-financial measures and multiple measures of performance?
 - they matter in their own right (e.g., product safety)
 - leading indicators of financial performance.
- Build them in – need to measure, evaluate, incentivize/compensate.
- What could go wrong?

Rules of the Game: How to Achieve Our Objectives

- Decide on ***objectives*** (owners).
- Design organizational **rules of the game** based on objectives:
 1. **Allocate decision rights** within the organization (which determines resource allocation),
 2. A **control system** that:
 - Measures and evaluates performance.
 - Ties compensation and non-monetary rewards to performance.

Cost concepts

- **Variable costs, fixed costs, and the relevant range** – costs are only fixed or variable within a defined range of activity and time period.
- **Contribution margin approach.** Know calculations and way of thinking about problems. Look at FinePrint; Zphone and broader applications (e.g., VMD, Birch Paper).
- **Opportunity costs, sunk costs.** Here again, concepts should be a way of thinking rather than specific calculations.

RegionFly

- *Perceived* profitability “too low”.
- Supply-driven “death spiral”
- Cut routes to improve profitability and did so based on routes (products) with “low” % margins.
- Allocated indirect costs using variable costs as activity measure – indirect costs declined more slowly than variable costs → higher allocation rate → more unprofitable routes.
- Why not focus on **dollars of contribution margin** (relative to fixed costs)?

[illegible]

		2015		2016		2017		2018	
Revenue		6,002,529		5,713,233		4,416,922		4,164,340	
Variable Costs:									
Variable Direct Costs		1,013,732	0.169	1,053,342	0.184	771,782	0.175	722,206	0.173
Flight Support (1000)		755,242	0.126	787,084	0.138	576,695	0.131	538,631	0.129
Maintenance (3000)		485,512	0.081	510,102	0.089	373,751	0.085	349,082	0.084
Airport expenses (7000)		242,756	0.040	252,165	0.044	184,761	0.042	172,566	0.041
		2,497,242		2,602,693		1,906,989		1,782,485	
Contribution Margin		3,505,287	0.584	3,110,540	0.544	2,509,933	0.568	2,381,855	0.572
Non-variable Costs:									
Sales expenses (2000)		134,865	0.022	138,900	0.024	117,983	0.027	111,485	0.027
Financing, fees, taxes (4000)		755,242	0.126	783,004	0.137	761,080	0.172	717,969	0.172
Depreciation (5000)		269,729	0.045	281,473	0.049	264,022	0.060	242,362	0.058
Admin costs (6000)		53,946	0.009	56,287	0.010	53,585	0.012	47,592	0.011
		1,213,782		1,259,664		1,196,670		1,119,408	
Net Profit		2,291,505	0.382	1,850,876	0.324	1,313,263	0.297	1,262,447	0.303

Was there anything wrong with how the business was doing in 2016?

Contribution Margins by Route for 2016

[illegible]

Contribution Margins by Route for 2016												
				1	2	3	4	5	6	7	Total	
	Revenues			1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233	
	Variable costs:											
	VDCs			167,593	195,632	210,962	107,596	130,971	79,338	161,250	1,053,342	
	1000			125,230	146,181	157,636	80,398	97,865	59,283	120,490	787,084	
	3000			81,160	94,739	102,163	52,106	63,425	38,421	78,089	510,102	
	7000			40,121	46,833	50,503	25,758	31,354	18,993	38,602	252,165	
				414,104	483,385	521,264	265,858	323,615	196,036	398,431	2,602,693	
	CM			634,693	346,629	632,555	257,695	484,511	277,196	477,261	3,110,540	
				0.61	0.42	0.55	0.49	0.60	0.59	0.55	0.54	
	Indirect non-variable overhead costs											
	2000										138,900	
	4000										783,004	
	5000										281,473	
	6000										56,287	
											1,259,664	
	Profit										1,850,876	

Consider pros and cons of allocating indirect overhead costs to routes; for reporting to board, for performance evaluation, for pricing, for resource allocation...

Contribution Margins by Route for 2016												
				1	2	3	4	5	6	7	Total	
	Revenues			1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233	
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	Profit										1,850,876	

Advocated a similar approach in evaluating profitability of different orders in MOP

VMD MIC

- Demand-driven “death spiral”
- Allocated indirect costs using DL hours.
 - Too much overhead allocated to labor-intensive products, too little to capital/tech-intensive products .
 - Systematic pricing errors when full costs used to set prices.

More granular cost allocation improved pricing decisions (if cost pools become more homogeneous).

- **Controllability principle** and buy-in by team.

Contribution Margin Approach										
			2015		2016		2017		2018	
Revenue			6,002,529		5,713,233		4,416,922		4,164,340	
Variable Costs:										
Variable Direct Costs			1,013,732	0.169	1,053,342	0.184	771,782	0.175	722,206	0.173
Flight Support (1000)			755,242	0.126	787,084	0.138	576,695	0.131	538,631	0.129
Maintenance (3000)			485,512	0.081	510,102	0.089	373,751	0.085	349,082	0.084
Airport expenses (7000)			242,756	0.040	252,165	0.044	184,761	0.042	172,566	0.041
			2,497,242		2,602,693		1,906,989		1,782,485	
Contribution Margin			3,505,287	0.584	3,110,540	0.544	2,509,933	0.568	2,381,855	0.572
Non-variable Costs:										
Sales expenses (2000)			134,865	0.022	138,900	0.024	117,983	0.027	111,485	0.027
Financing, fees, taxes (4000)			755,242	0.126	783,004	0.137	761,080	0.172	717,969	0.172
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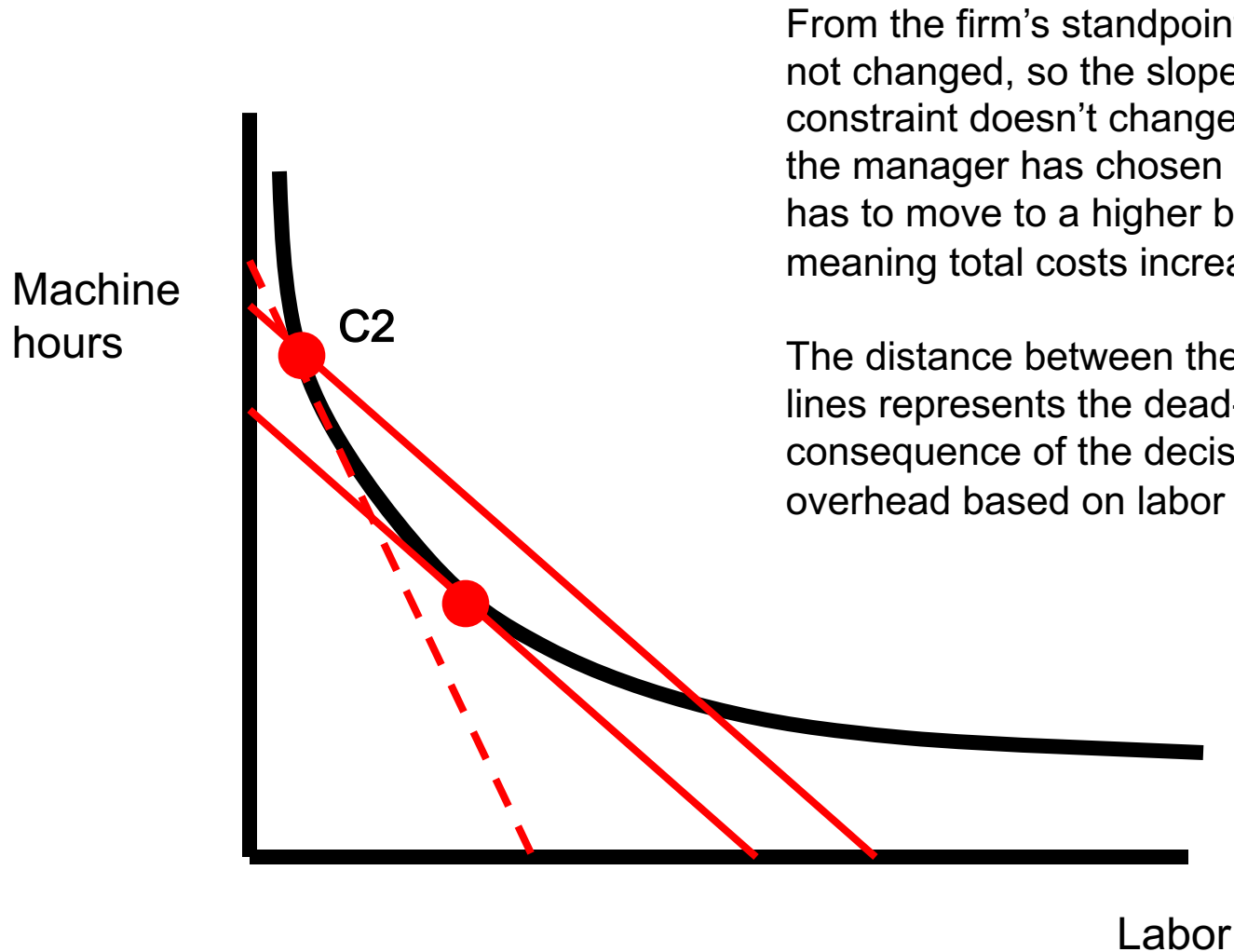


Whenever we have indirect costs (including joint costs), any allocation is *arbitrary* in the sense that there is no single “correct” allocation (and so no “true” cost). Allocations then must be fit for purpose.

(*Joint costs* are the costs of a production process that yields multiple products simultaneously. For example, refining a barrel of crude oil simultaneously produces gasoline, diesel fuel, asphalt, lubricants, etc.)

So why allocate costs?

1. Set prices within the organization; necessary for resource allocation:
 - Internal price of shared/common resources should not be zero.
 - Changing internal prices (by allocating indirect costs or doing so in different ways) affects decisions.
 - Can we measure the *marginal cost* of resource consumption and use this to set internal prices? Average cost may be “good enough”.
 - Transfer prices are used for this purpose.

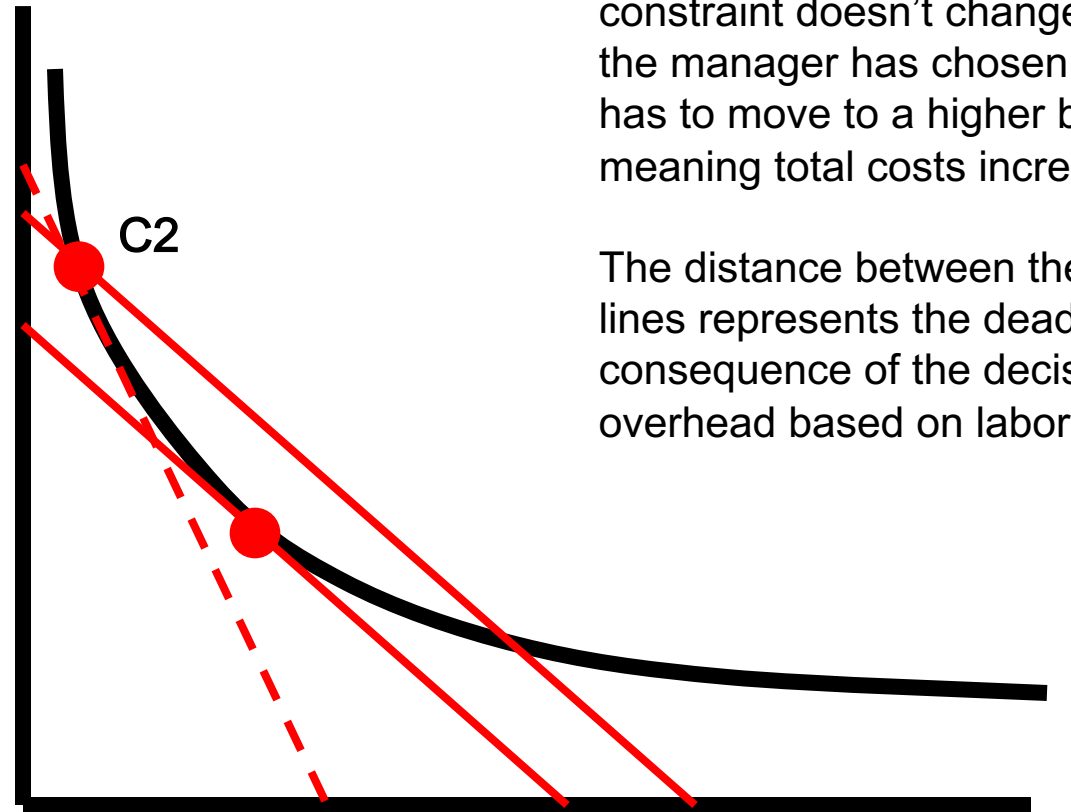


From the firm's standpoint, prices have not changed, so the slope of the budget constraint doesn't change. But because the manager has chosen C2, the firm has to move to a higher budget line, meaning total costs increase.

The distance between the two solid lines represents the dead-weight loss, a consequence of the decision to allocate overhead based on labor hours.

Analogous to what we saw in Birch Paper – if we decentralize, outcome may not optimize overall organizational value (we can't get to the "first best" solution).

Machine
hours



From the firm's standpoint, prices have not changed, so the slope of the budget constraint doesn't change. But because the manager has chosen C2, the firm has to move to a higher budget line, meaning total costs increase.

The distance between the two solid red lines represents the dead-weight loss, a consequence of the decision to allocate overhead based on labor hours.

Labor

So why allocate costs?

2. GAAP and tax rules require overhead cost allocation in manufacturing: **Absorption costing.**

Understand Bullwinkle example and how absorption costing can be used to manage reported gross margins on the income statement.

Also necessary for regulatory purposes – e.g., in investment management industry, investment complexes allocate overhead costs (e.g., audit fees for fund complex) to individual mutual fund products which is necessary to evaluate reasonableness of fees charged by investment managers.

Absorption Costing and Manufacturing Overhead

Assume \$2M of fixed manufacturing overhead cost for 2020 and no opening inventory

Management sets Normal Capacity = 100,000

FMOH/unit = \$20.00

Assume also:

Production for 2020 80,000

Sales for 2020 75,000

What's reported in COGS and Closing Inventory?

Closing inventory = **\$100,000** (Production - Sales) x FMOH/unit

COGS (allocated to product) \$1,500,000 Sales x FMOH/unit

Remaining unallocated FMOH \$400,000

Total COGS = **\$1,900,000**

Total FMOH **\$2,000,000**

(Total FMOH must be fully allocated to COGS + Closing Inventory)

Summary of inventory costing under absorption costing vs. for internal purposes:

Components of inventory costs:	Absorption costing (GAAP)	Internal decisions, performance evaluation, control
Fixed MOH/unit	Computed using Normal Capacity	Do not allocate
Variable MOH/unit	Computed using Actual Production	Compute using Practical Capacity or do not allocate.
Direct material and Direct Labor per unit	Include	Include
Total Unit Costs	Inventory cost/unit for Financial Reporting	Inventory cost/unit for internal decision-making and control

$$\text{Cost of Resources Supplied} = \text{Cost of Resources Used} + \text{Cost of Unused Capacity}$$

↑
Allocate to
activity/process
using practical
capacity.

↑
Don't allocate to
product; use for
control

Tesla operates a plant with practical capacity of 200,000 units per year. The cost of supplying capacity (keeping the plant running to produce vehicles) is \$50M/year, a fixed cost each year.

Over the first three years of plant operations it produces 50K, 70K, and 100K vehicles, respectively. How should we allocate the cost of supplying capacity?

Year 1: $\$50\text{M}/50\text{K} = \$1,000$ per unit.

Year 2: $\$50\text{M}/70\text{K} = \714 per unit.

Year 3: $\$50\text{M}/100\text{K} = \500 per unit.

Tesla operates a plant with *practical capacity* of 200,000 units per year (max is 225K). The cost of supplying capacity (keeping the plant running to produce vehicles) is \$50M/year, a fixed cost each year.

Over the first three years of plant operations it produces 50K, 70K, and 100K vehicles, respectively. How should we allocate the cost of supplying capacity?

Year 1: $\$50\text{M}/50\text{K} = \$1,000$ per unit.

Year 2: $\$50\text{M}/70\text{K} = \714 per unit.

Year 3: $\$50\text{M}/100\text{K} = \500 per unit.

This is the wrong answer – we are *allocating all of the \$50M to vehicles produced*, and the cost system is telling us that we are getting more efficient. And if production levels decline, more indirect cost is allocated to each vehicle, leading to the death spiral scenario.

		Allocated to vehicles	Cost of Unused Capacity	Total indirect cost of capacity
Year 1	\$250 x 50K	\$12.5M	\$37.5M	\$50M
Year 2	\$250 x 70K	\$17.5M	\$32.5M	\$50M
Year 3	\$250 x 100K	\$25M	\$25M	\$50M
Based on practical capacity of 200K vehicles/year we have $\$50\text{M}/200\text{K} = \250 allocated to each vehicle. This rate stays constant (assuming the production process is unchanged).				

We use this same principle to implement ABC (e.g., practical capacity of delivery drivers).

So why allocate costs?

3. Cost-plus or full-cost reimbursement contracts.

In the long run the organization has to **recover all costs** from customers to be profitable and survive.

This **includes sunk costs**, such as past CAPEX (which is why depreciation is included in overhead cost pools).

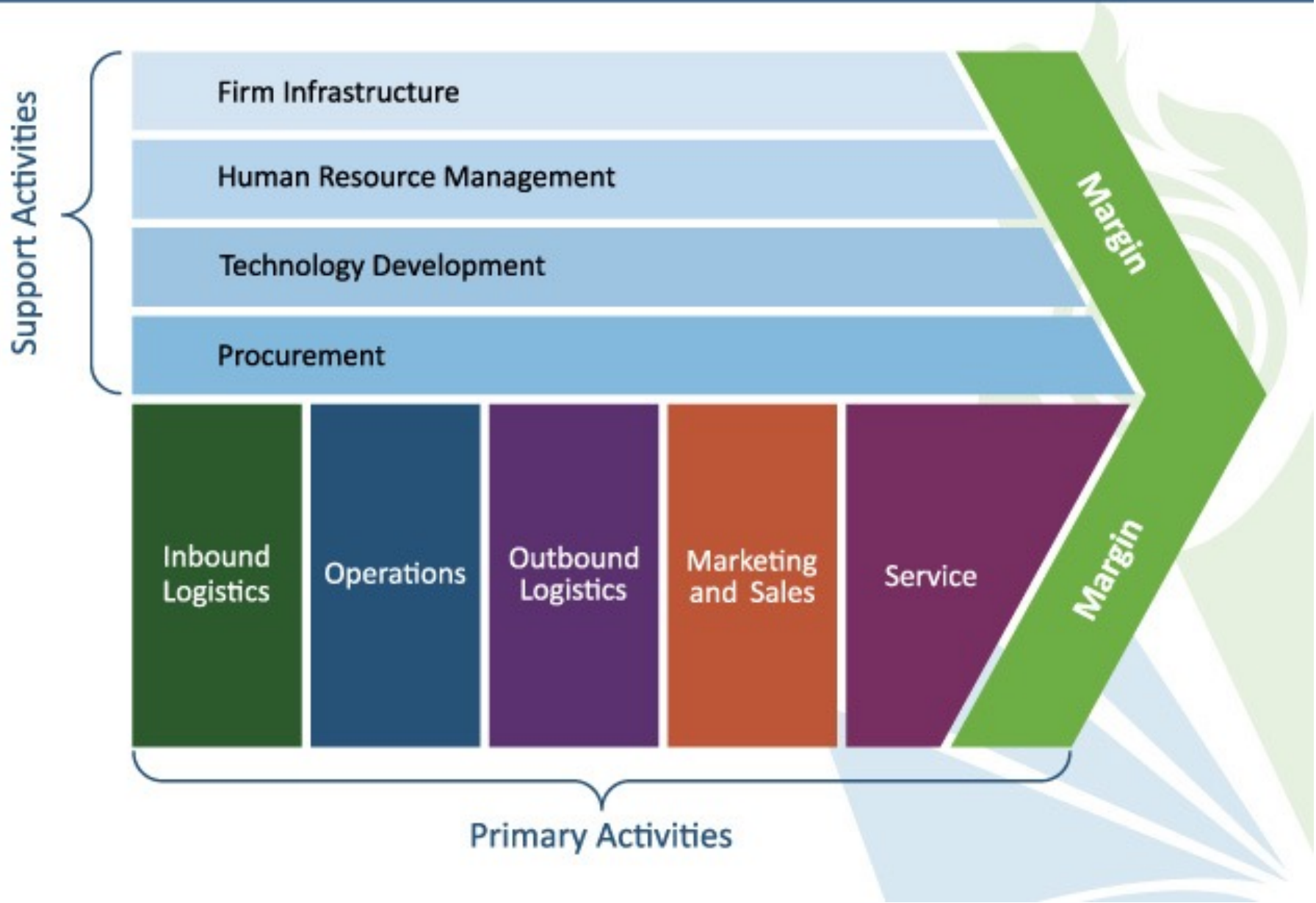
Given complications of allocating indirect overhead/support costs to business units, perhaps use a *contribution margin* approach (i.e., revenues – variable direct costs)?

Two main issues:

- How does organization decide how much to invest in the support function (i.e., its size)?
- How does organization decide how to allocate these scarce resources?

For example, how large a team do we need at Booth to provide administrative support for faculty?

Value Chain Analysis



**“Why did I rob banks?
Because I enjoyed it. I loved it.
I was more alive when I was
inside a bank, robbing it, than
at any other time in my life.”**

**“I enjoyed everything about it
so much that one or two
weeks later I'd be out looking
for the next job. But to me the
money was the chips, that's
all!”**



So Willie didn't really rob banks because "that's where the money is"

ABC: Where to look:

- Areas where there are **large chunks of costs** – start with the Income Statement.
- Look for resource consumption that increases with **diversity/complexity** of products, processes, or customers.
- Look for resources whose consumption is not well explained by conventional drivers like units produced.
- Look over periods of years, not months, to identify **longer-run, underlying trends** in cost structure as business evolves.

Birch Paper

- Know pros and cons of the key alternatives:
 - Allow Northern to make its own decision. Preserves integrity of decentralized system but doesn't maximize Birch profits.
 - Will Brunner conclude \$429 is the “right answer” and adjust his offer, or will he insist on \$480?
 - Use a dual price (Thompson sells @ \$480, Northern buys @ \$430). Maximizes Birch profits and keeps Kenton and Brunner happy but requires intervention.
 - *What will happen as the game repeats over time?*

Hong Kong / Transport

Hong Kong's Cathay Pacific Airways warns pilots against slow taxiing after some accused of congesting airport

- Internal airline memo seen by Post says some pilots going at 'considerably slower speed' while taxiing
- Union leader says go-slow issues reflect poor morale among pilots as salaries stay at 60 per cent of pre-pandemic levels despite record profits

 Ezra Cheung

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Published: 7:55pm, 14 May, 2023 ▾

 [Why you can trust SCMP](#)

Source: South China Morning Post, May 14, 2023.

Regardless of what top management might decide is optimal, nothing matters unless people (1) understand, and (2) buy-in to how their performance is measured and evaluated. Participation and communication is key.

Hong Kong / Transport

Hong Kong's Cathay Pacific Airways warns pilots against slow taxiing after some accused of congesting airport

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More on Rules of the Game

- Understand pros and cons of different responsibility center structures (cost centers, profit centers, etc.).
- Budgets used to both set goals and expectations as well as for performance measurement and evaluation.
- Know variance definition.

P = Actual Price

Q = Actual Quantity

EP = Budgeted Price

EQ = Budgeted Quantity

Total variance = $P.Q - (EP.EQ)$

= $[Q.(P - EP)] + [EP.(Q - EQ)]$

= Price variance + Quantity variance

Applies to revenues and costs.

Please stay in touch!

dskinner@chicagobooth.edu

Thank you!!!